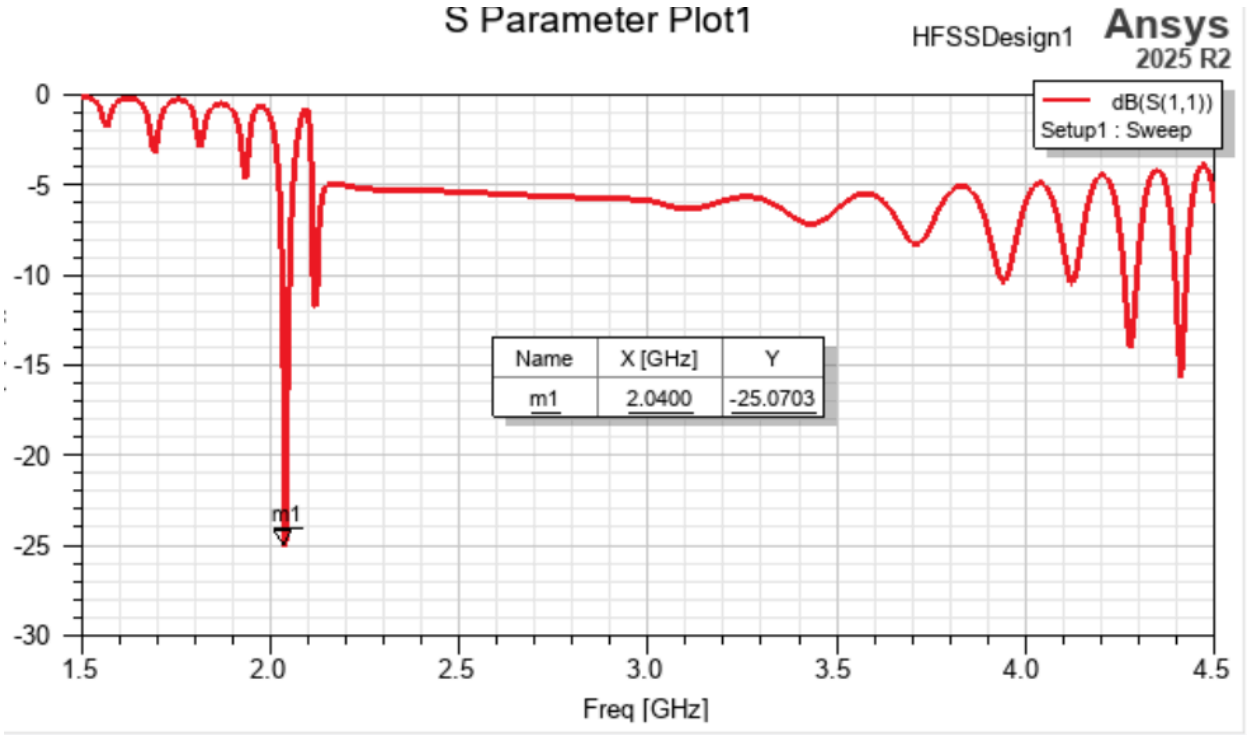
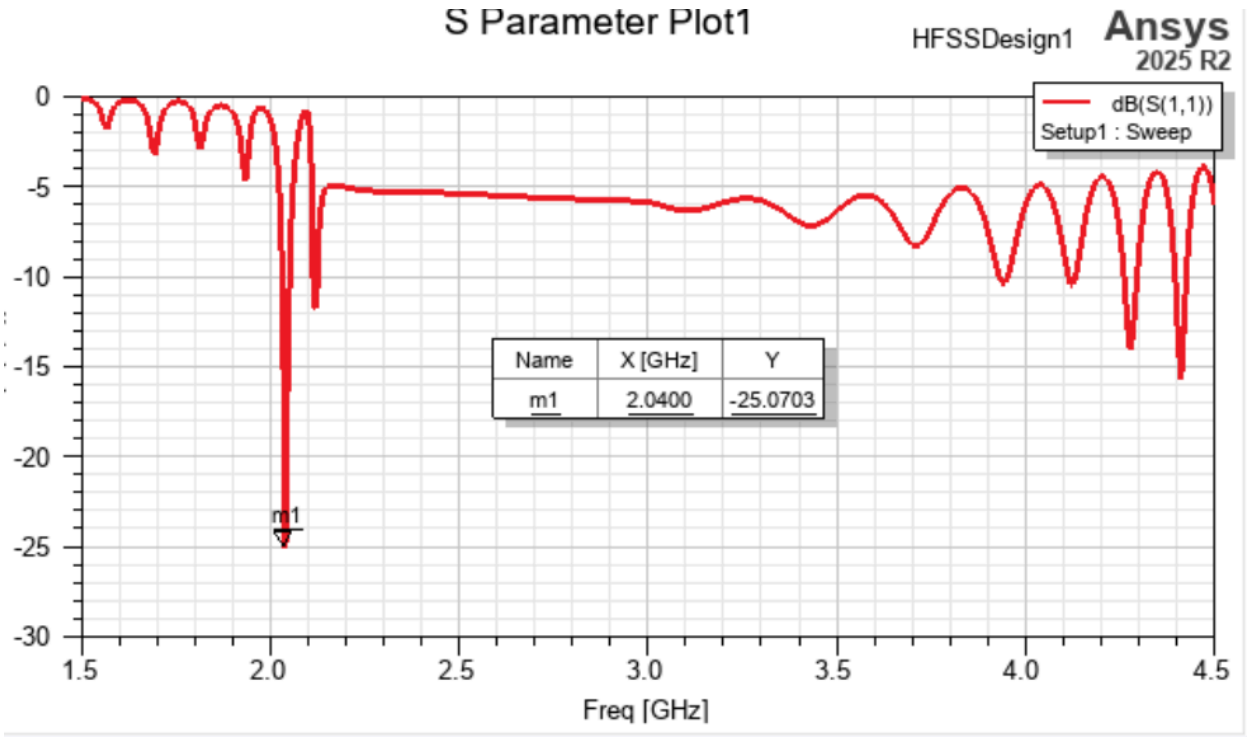
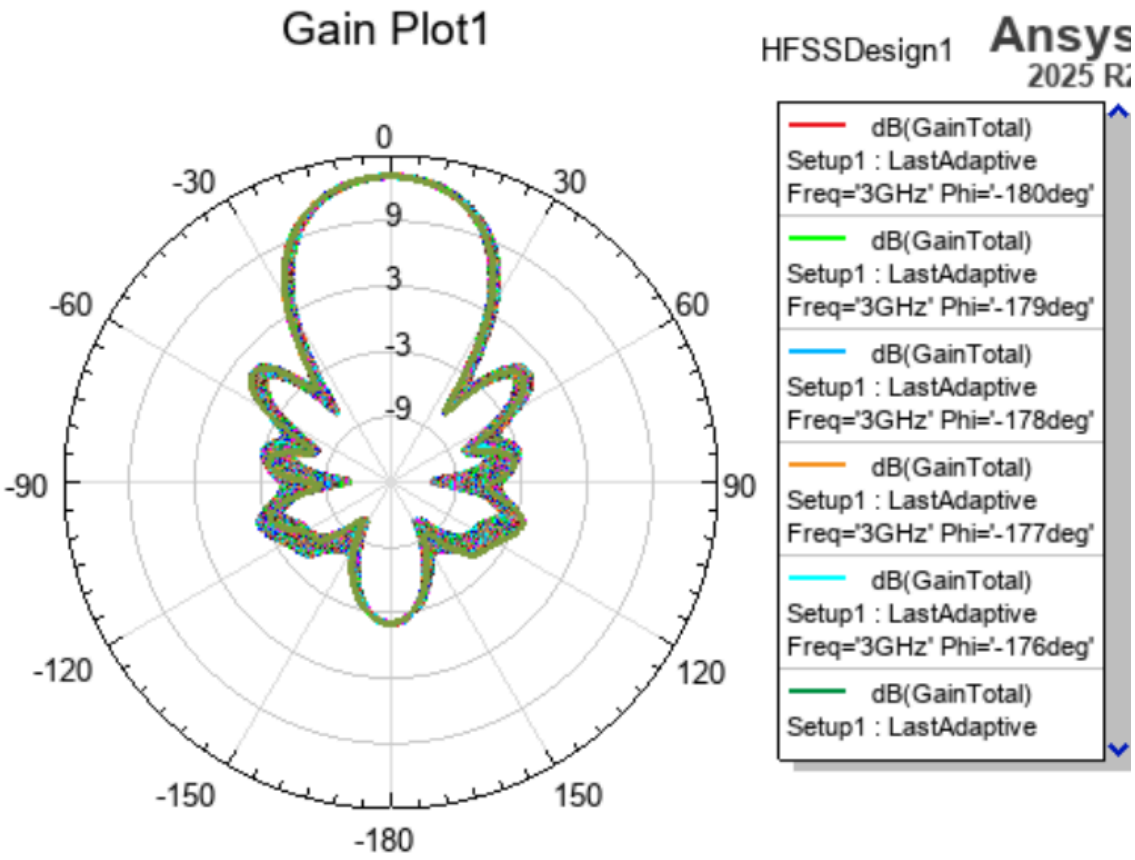
	Freq [GHz]	dB(PeakGain) Setup1 : LastAdaptive
1	3.000000	13.088400

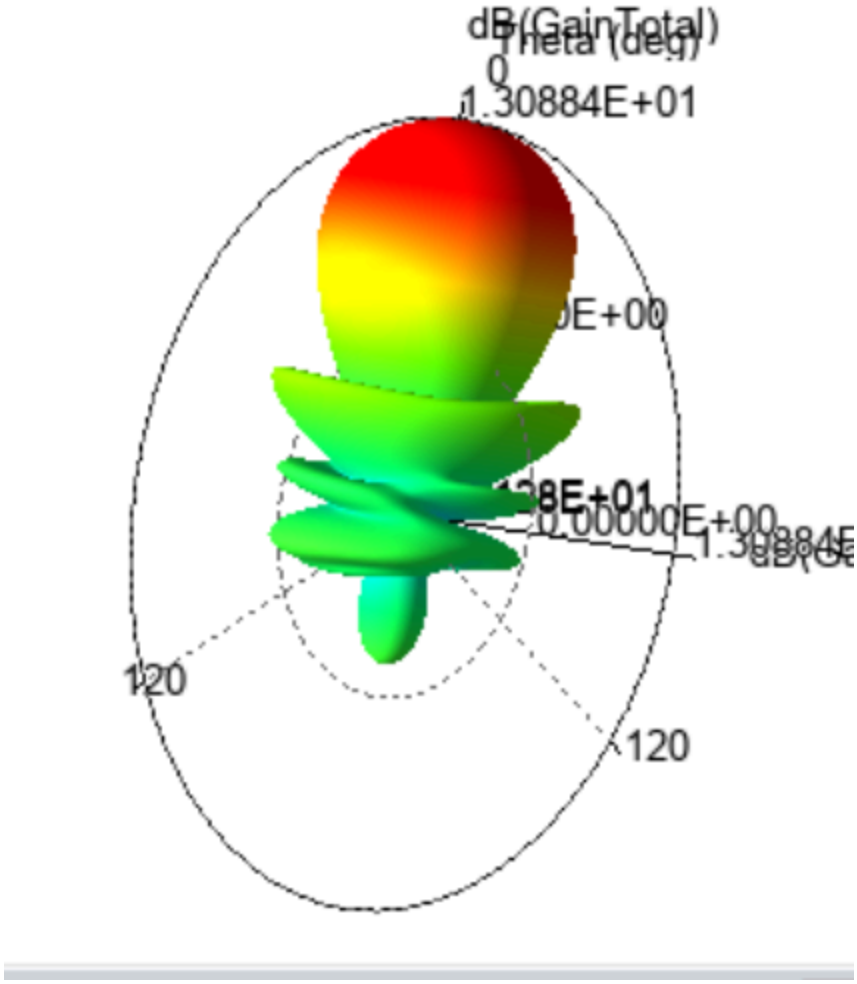
	Freq [GHz]	dB(PeakGain) Setup1 : LastAdaptive
1	3.000000	13.088400



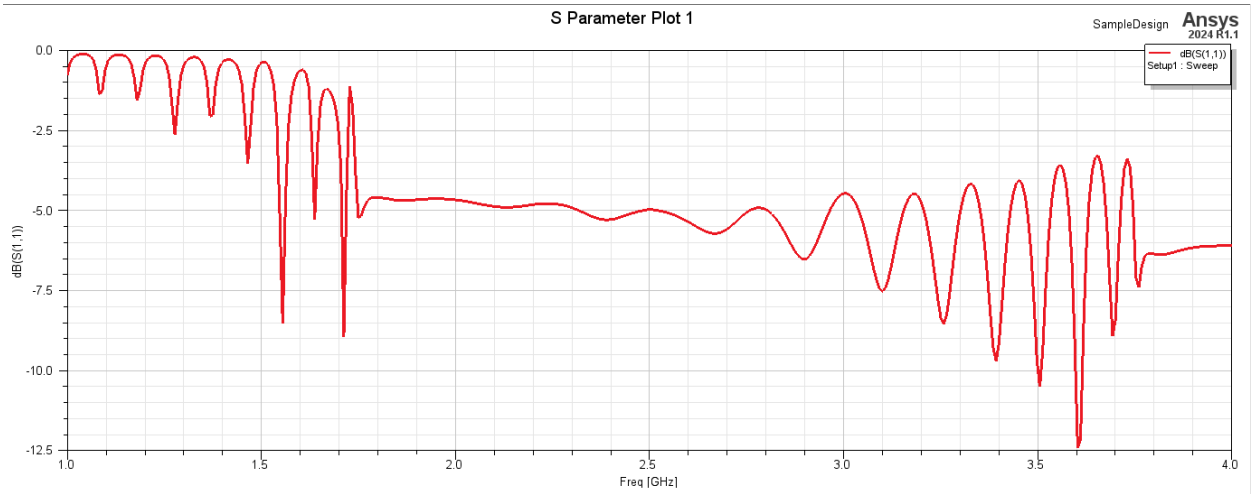

S Parameter Plot1
HFSSDesign1
Ansys
2025 R2
dB(S(1,1))
Setup1 : Sweep
Name
X [GHz]
Y
m1
2.0400
-25.0703
m1
0
-5
-10
-15
-20
-25
-30
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Freq [GHz]


Gain Plot1
HFSSDesign1
Ansys
2025 R
0
30
60
90
120
150
-180
-150
-120
-90
-60
-30
9
3
-3
-9
dB(GainTotal)
Setup1 : LastAdaptive
Freq='3GHz' Phi='-180deg'
dB(GainTotal)
Setup1 : LastAdaptive
Freq='3GHz' Phi='-179deg'
dB(GainTotal)
Setup1 : LastAdaptive
Freq='3GHz' Phi='-178deg'
dB(GainTotal)
Setup1 : LastAdaptive
Freq='3GHz' Phi='-177deg'
dB(GainTotal)
Setup1 : LastAdaptive
Freq='3GHz' Phi='-176deg'
dB(GainTotal)
Setup1 : LastAdaptive

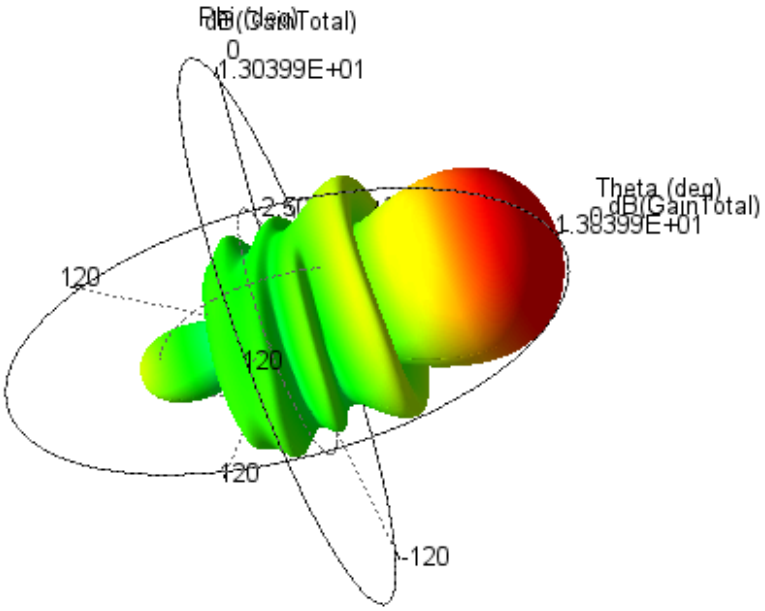
Gain Plot2


dB(GainTotal)
Theta (deg)
0
1.30884E+01
E+00
8E+01
0.00000E+00
120
120

Now lets tune for 2.4 GHz…



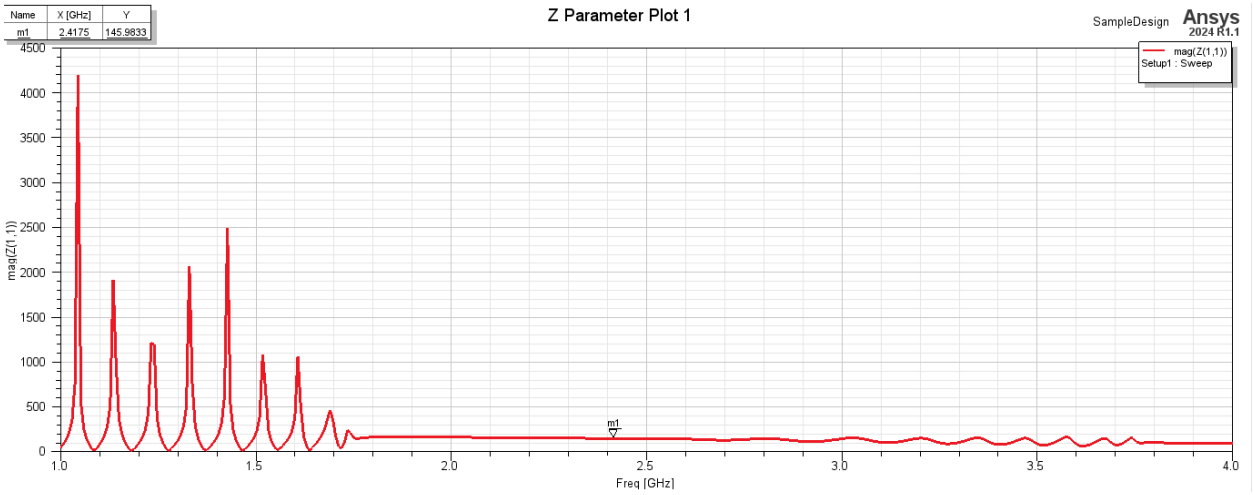
Shit



	Freq [GHz]	dB(PeakGain) Setup1 : LastAdaptive
1	2.400000	13.039946

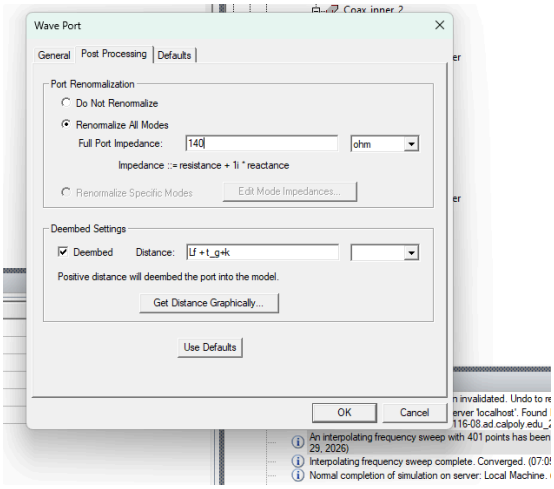
Not bad though…

Deembedding ports to figure out impedance of antenna for proper matching.



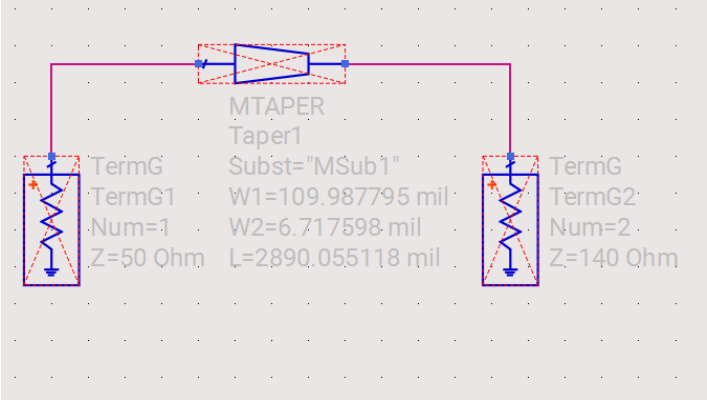
Well well well….

Going to renormalize the port impedance to 140 ohms→



Impedance Transformer

The typical impedance of an axial mode helical antenna is around 140 Ohms, there needs to be some kind of impedance transformation between the 50 SMA connection and 140 Ohms of the antenna. Tapered line yields traces that are too small.



SOLUTION: SINGLE QUARTER WAVE IMPEDANCE TRANSFORMER

Each length of the section is

$$l_i = \frac{c}{4f_o\sqrt{\epsilon_{eff,i}}}$$

and the width is

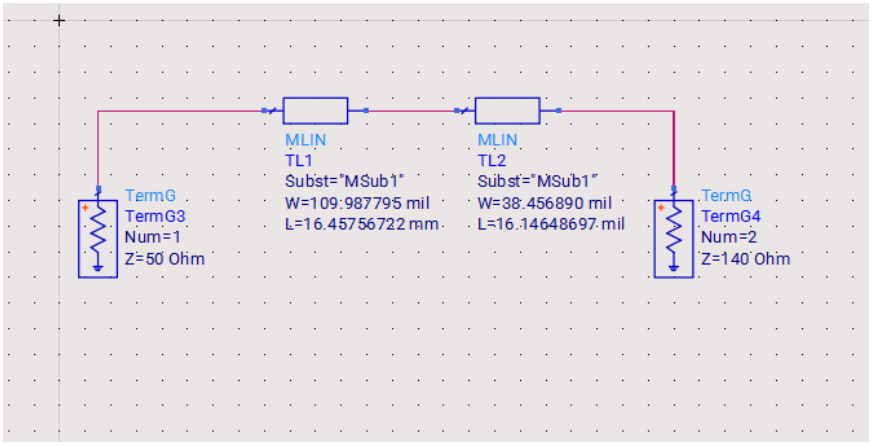
$$Z_{0,\frac{\lambda}{4}} = \sqrt{Z_{in}Z_L}$$

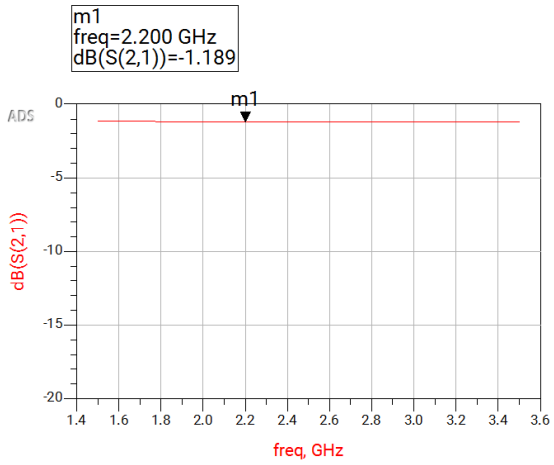
where Zin should be 50 Ohms to match the input impedance of the SMA.

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}\left(1 + 12\frac{h}{w}\right)^{-\frac{1}{2}}$$

Effective permittivity per balanis's textbook^.

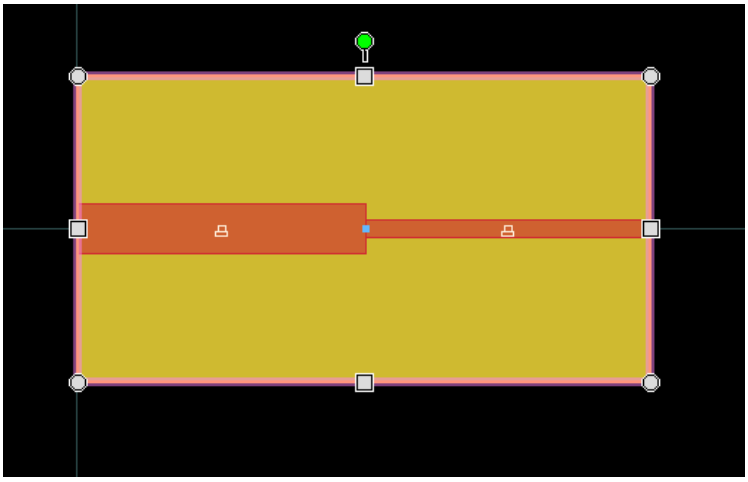
Here is the result in ADS:

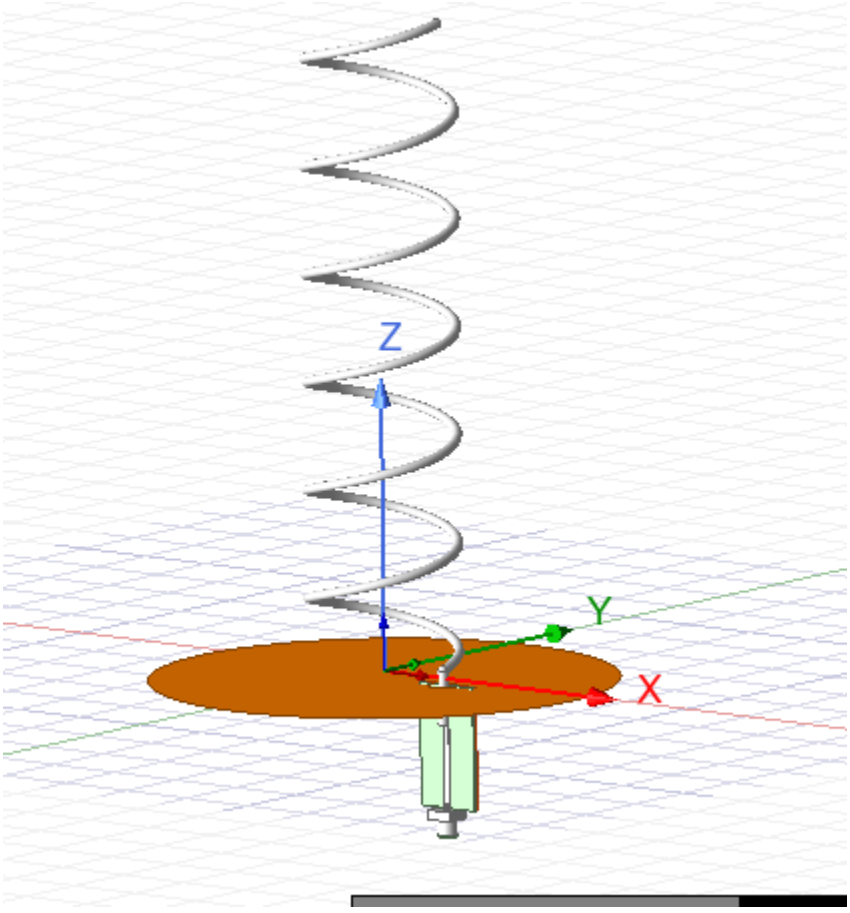




-About a dB of insertion loss at the center frequency. Good!!

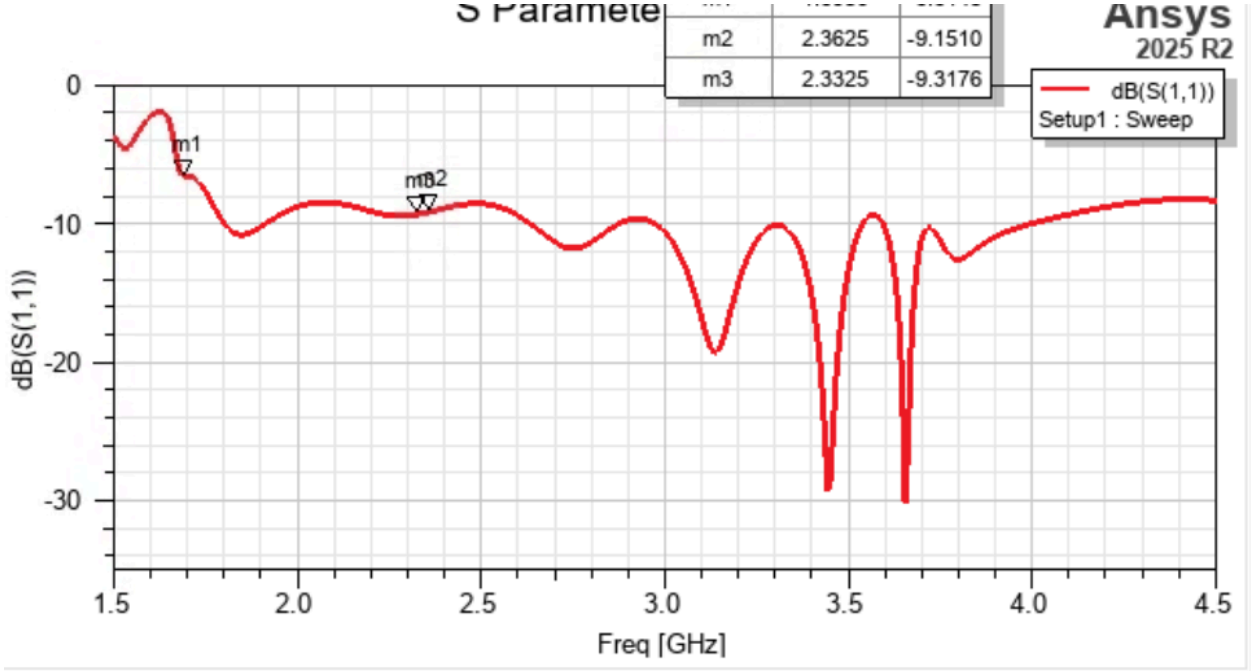
Layout:





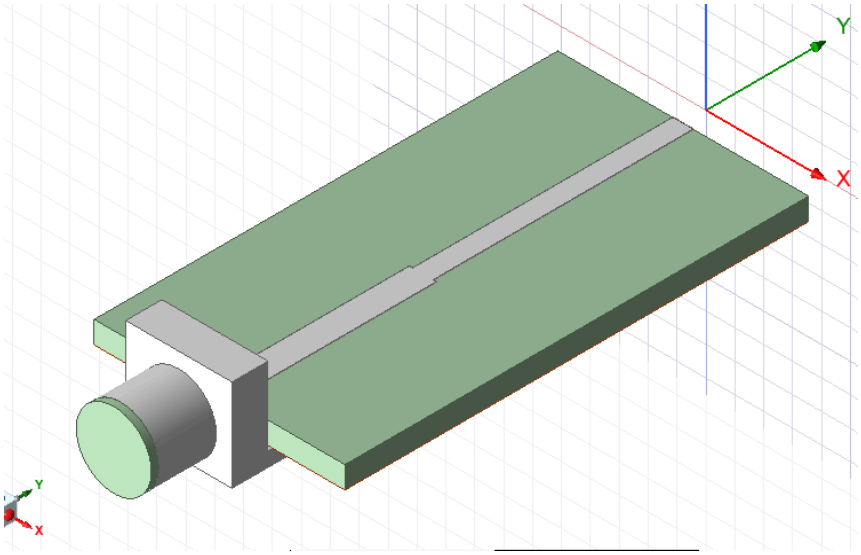
Antenna with quarter wavelength transformer

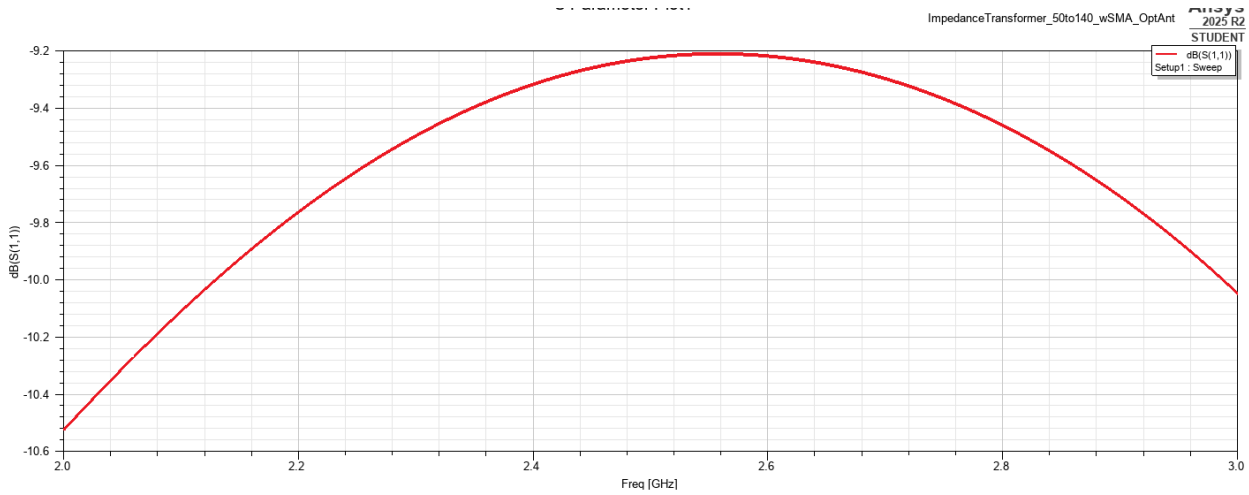
Results:



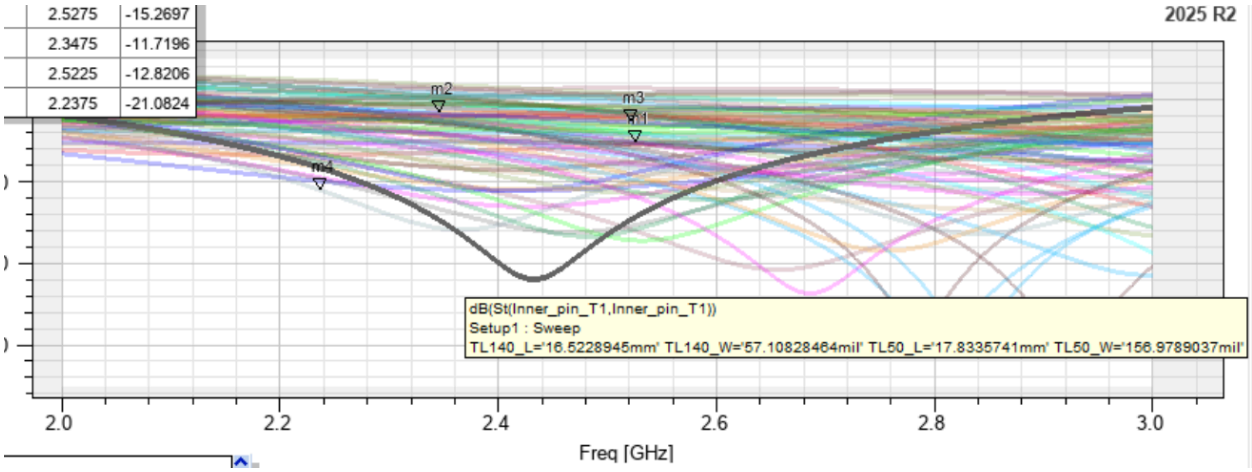
Pretty terrible ngl

Reinvestigating the quarter wavelength transformer:





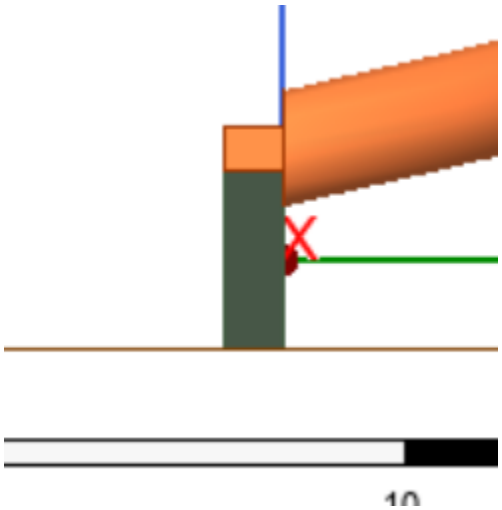
Bad when the top sheet is thickened



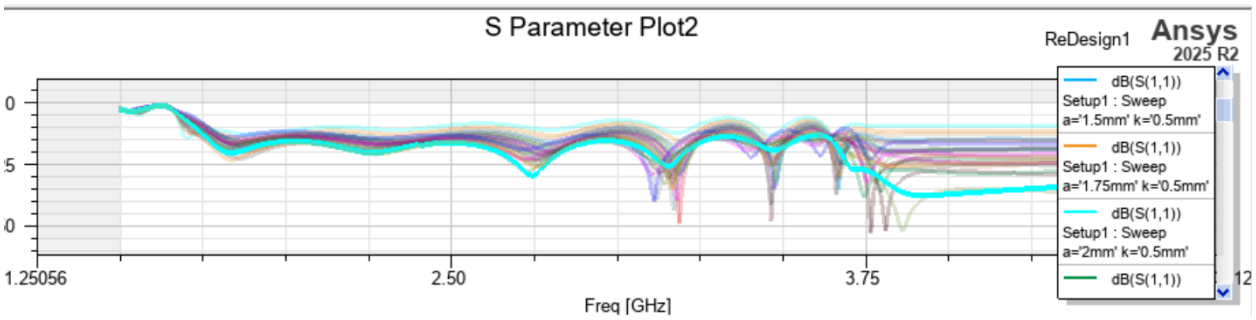
Initial optimization result of QWT

Best result

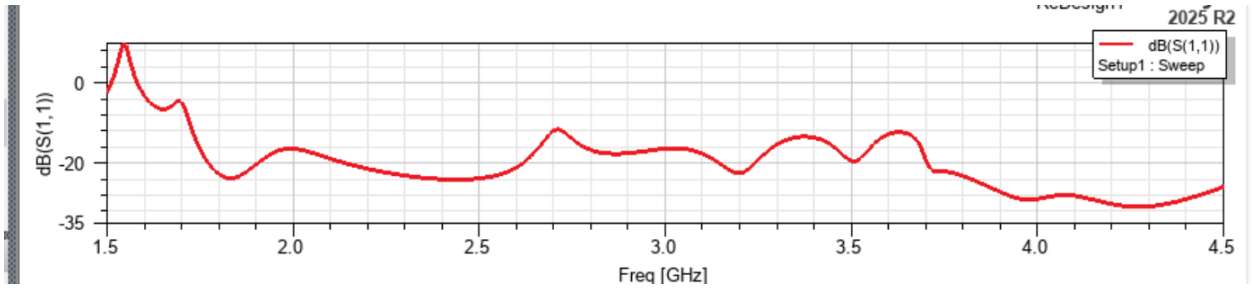
Changing the antenna to better match this broadband design in this paper:
Changing the feed to do an impedance study



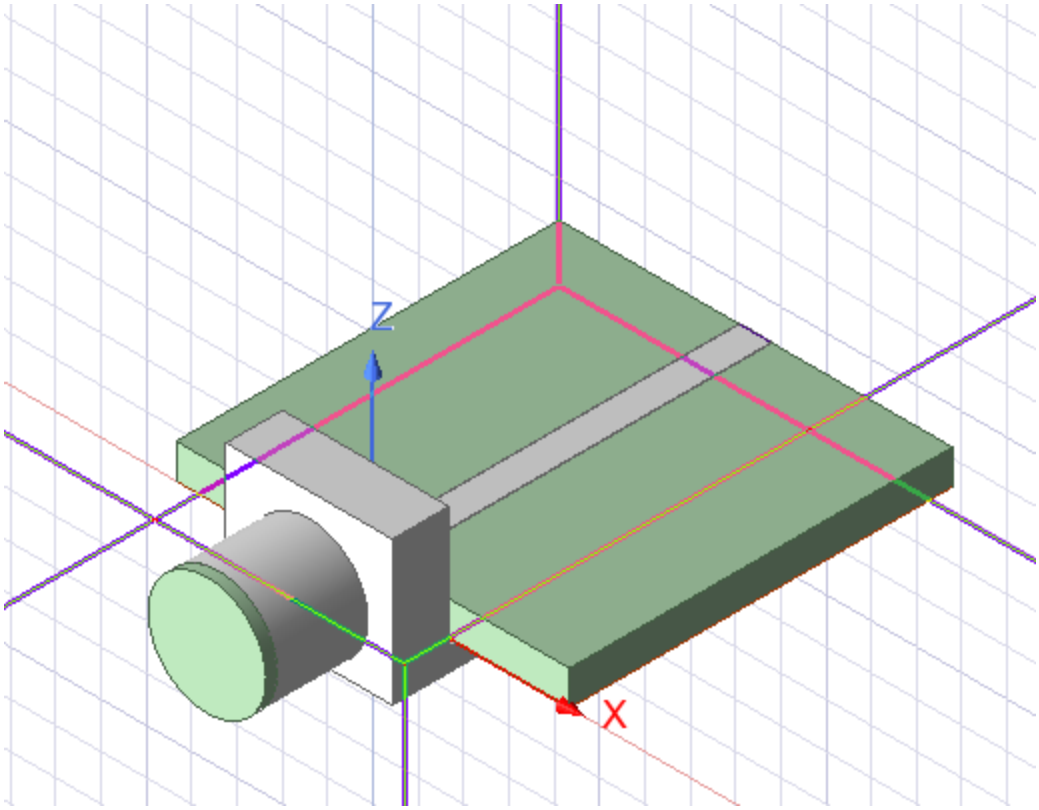
Going to redesign impedance transformer
ZL = 70.71067812

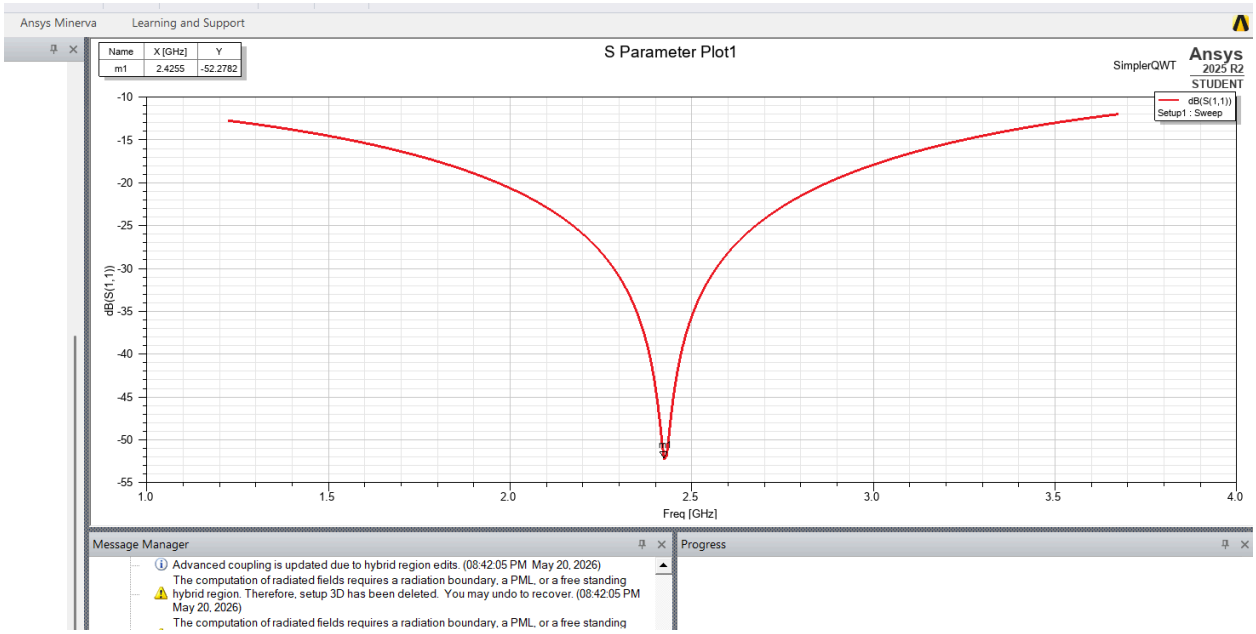


Mann

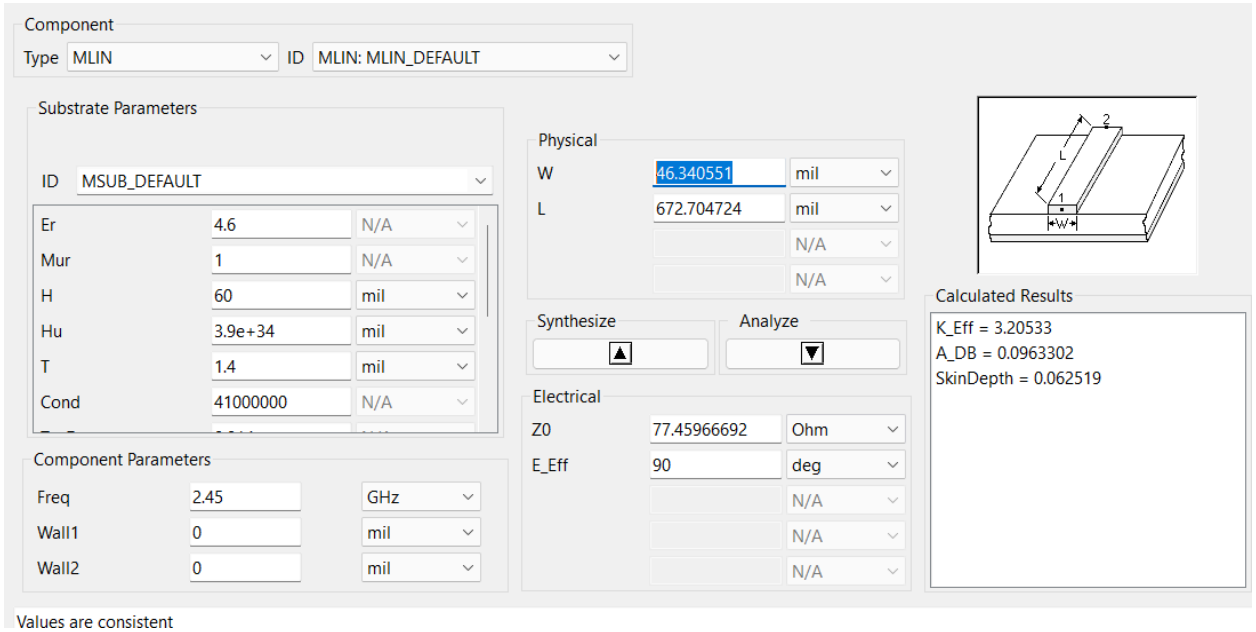


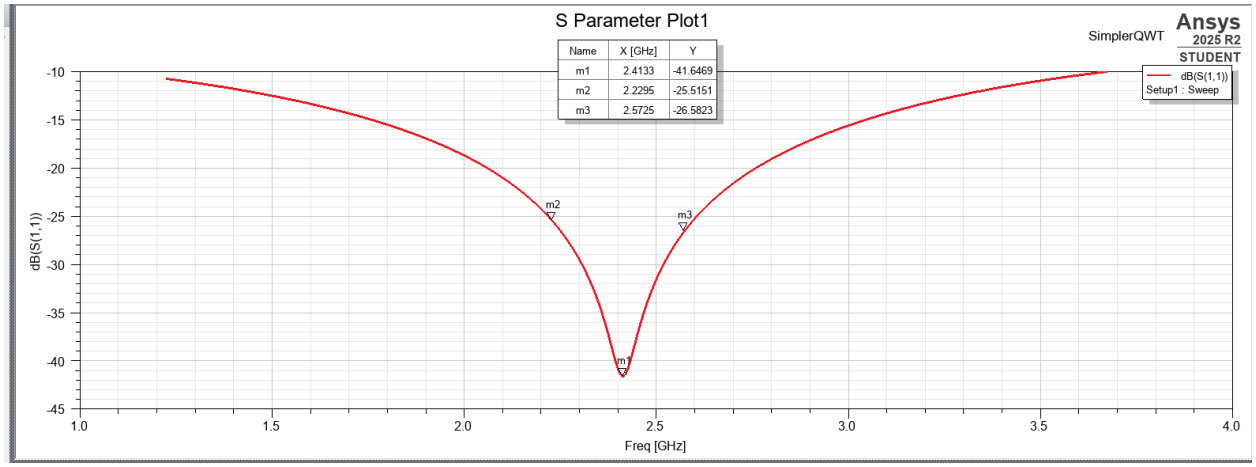
With 120 ohm lumped port



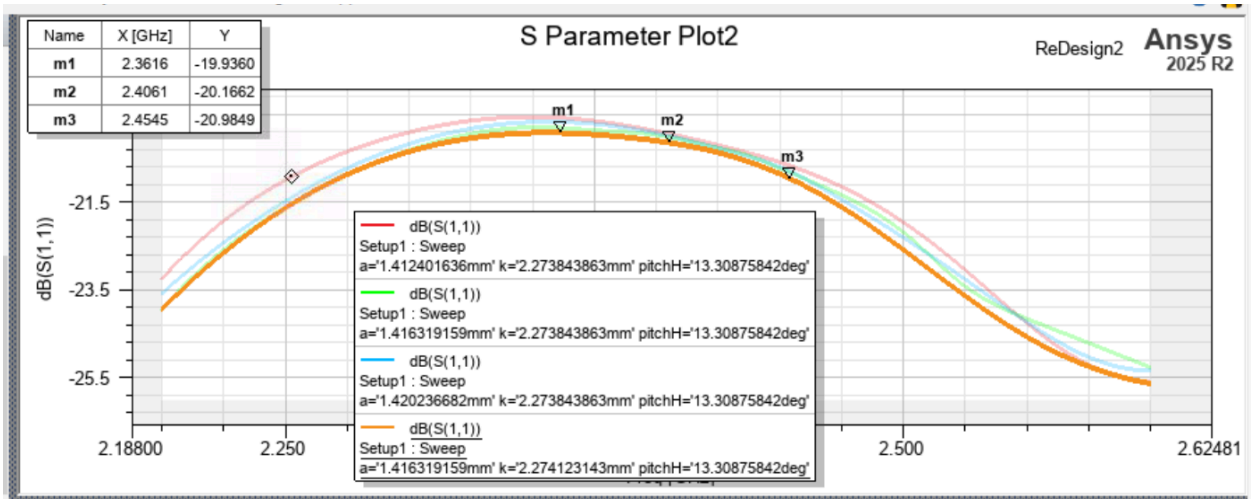


Resizing QWT for 120 ohms

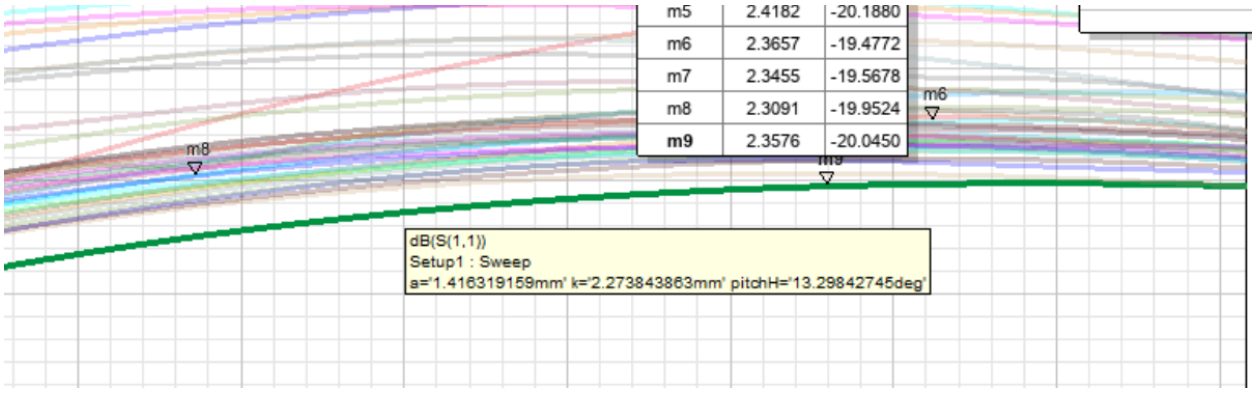

Component
Type MLIN
ID MLIN: MLIN_DEFAULT
Substrate Parameters
ID MSUB_DEFAULT
Er 4.6 N/A
Mur 1 N/A
H 60 mil
Hu 3.9e+34 mil
T 1.4 mil
Cond 41000000 N/A
Component Parameters
Freq 2.45 GHz
Wall1 0 mil
Wall2 0 mil
Physical
W 46.340551 mil
L 672.704724 mil
N/A
N/A
Synthesize
Analyze
Electrical
Z0 77.45966692 Ohm
E_Eff 90 deg
N/A
N/A
N/A
Calculated Results
K_Eff = 3.20533
A_DB = 0.0963302
SkinDepth = 0.062519
Values are consistent


S Parameter Plot1
SimplerQWT
Ansys
2025 R2
STUDENT
Name X [GHz] Y
m1 2.4133 -41.6469
m2 2.2295 -25.5151
m3 2.5725 -26.5823
dB(S(1,1))
Setup1 : Sweep
dB(S(1,1))
Freq [GHz]

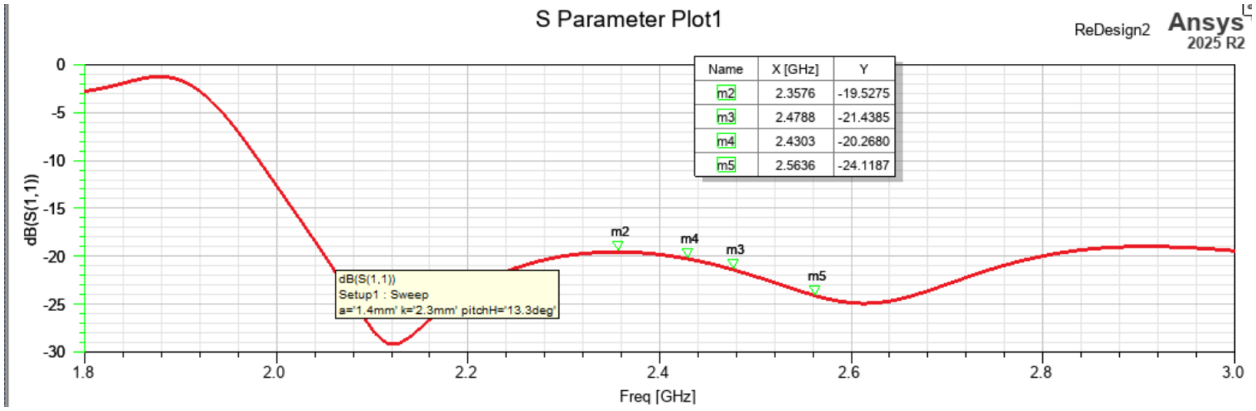
Still works



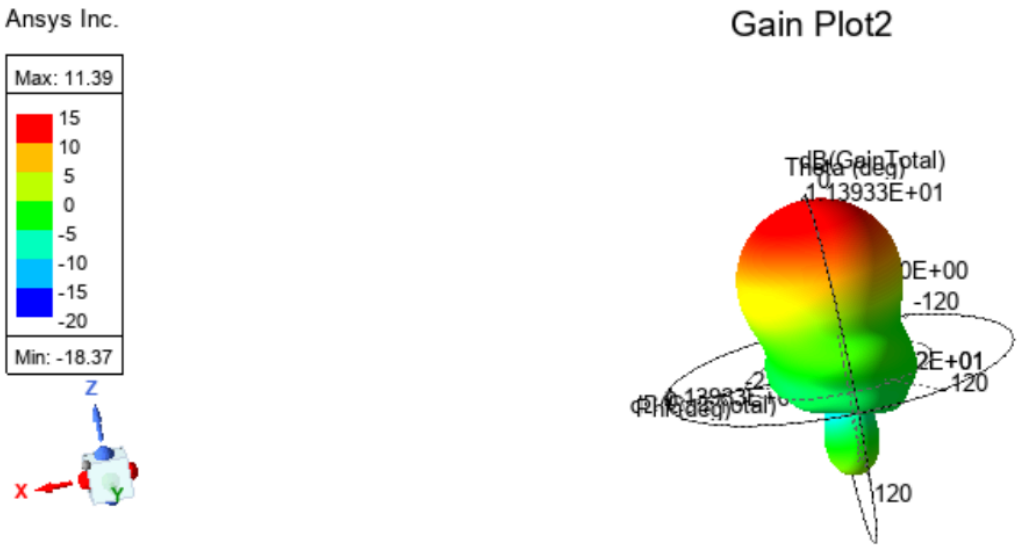
Helical antenna optimization result, probably going to stop here.



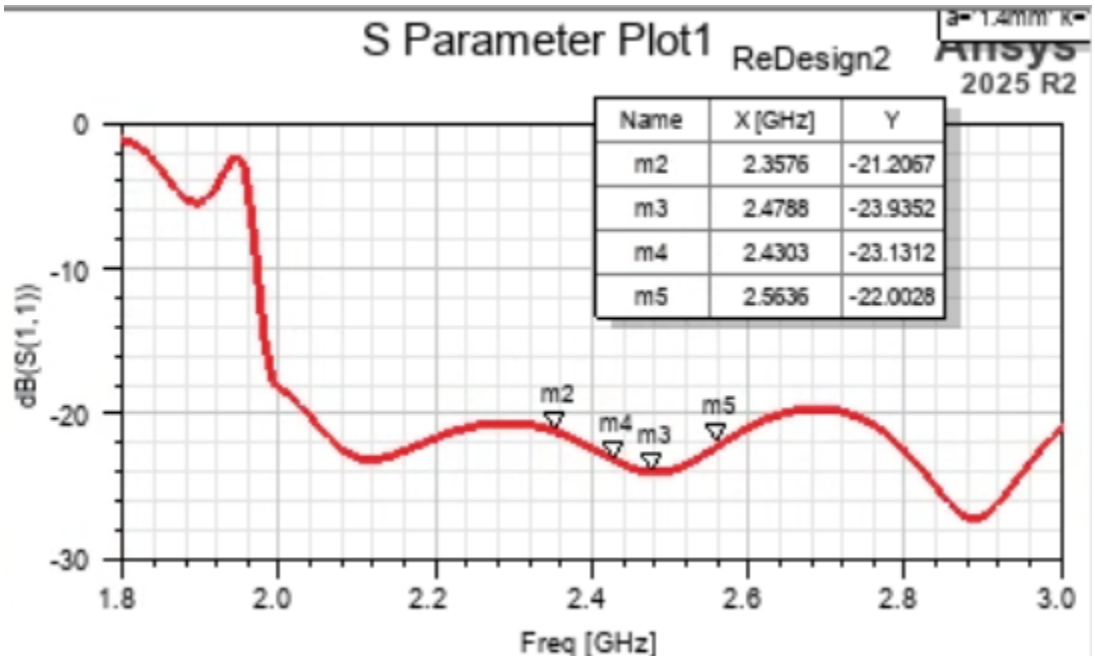
Going to go with this one here



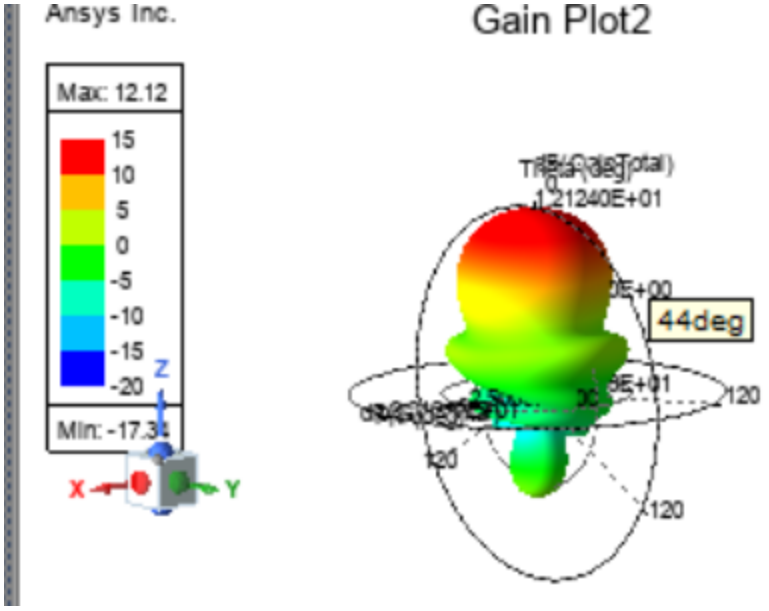
Omitted some of the precision, the chosen parameters for the helical antenna are shown on the plot, wire radius of 1.4mm, helix offset of 2.3 mm , and pitch of 13.3 mm



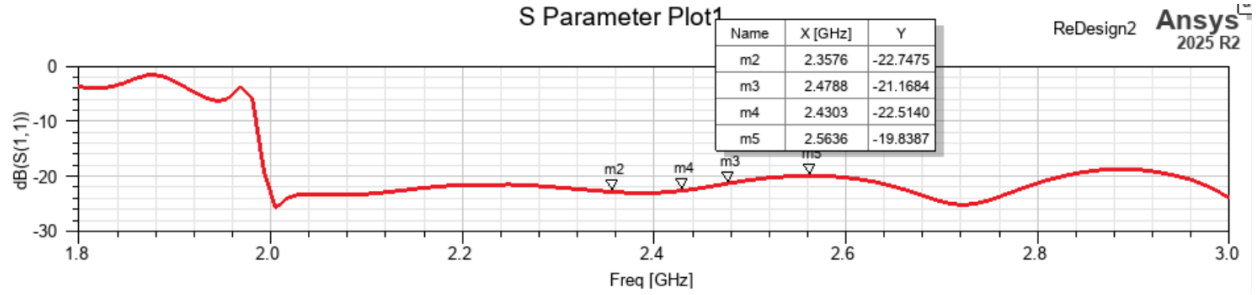
Fairly decent gain, going to add more turns.

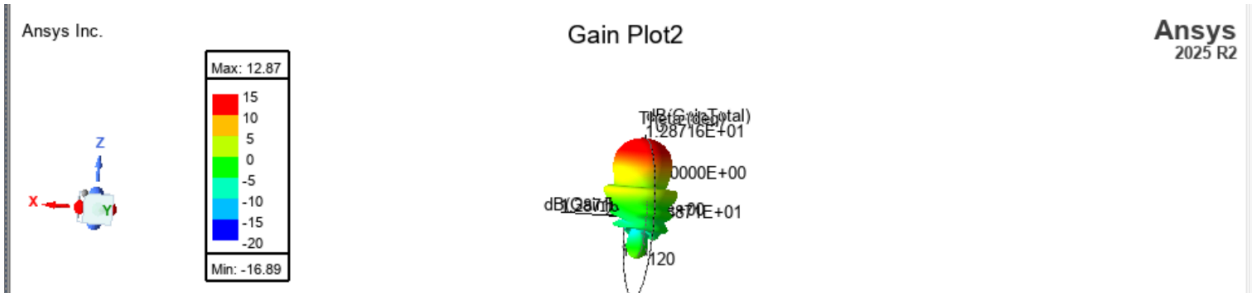


Gain and matching looks much better with 9 turns.



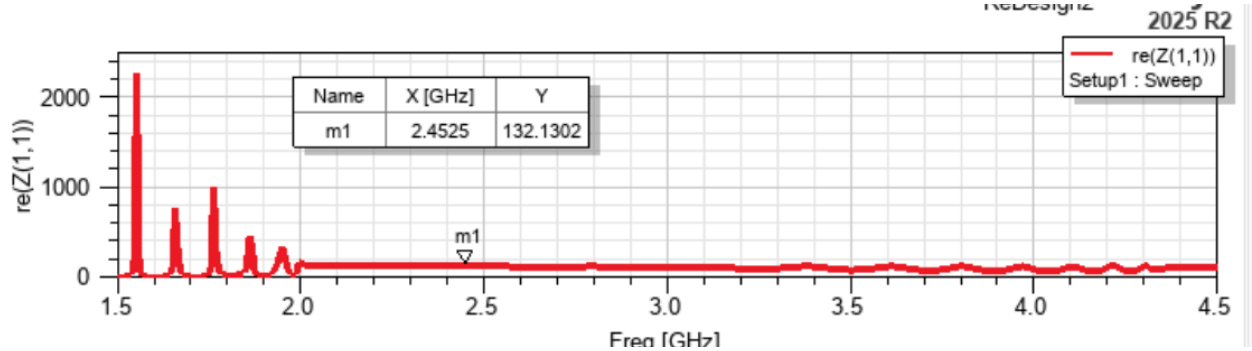
Going to try adding 12 turns

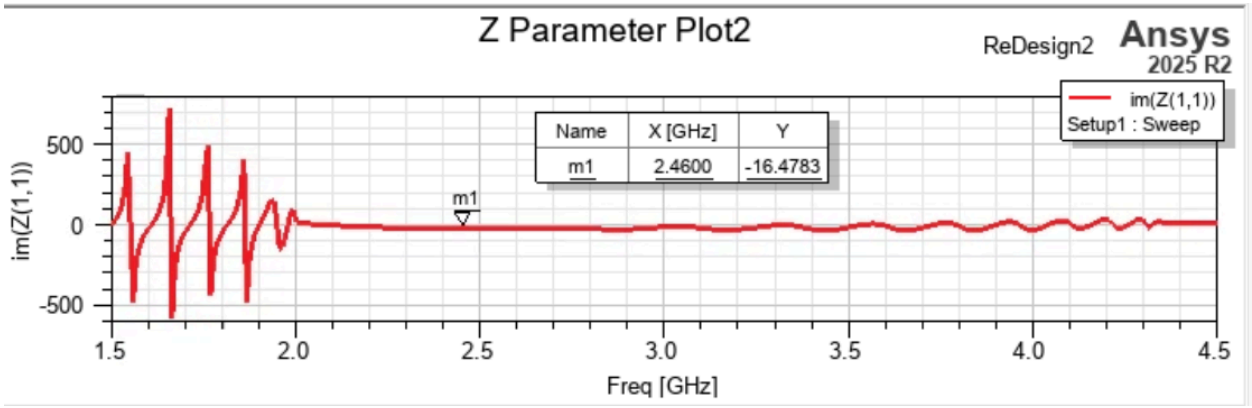


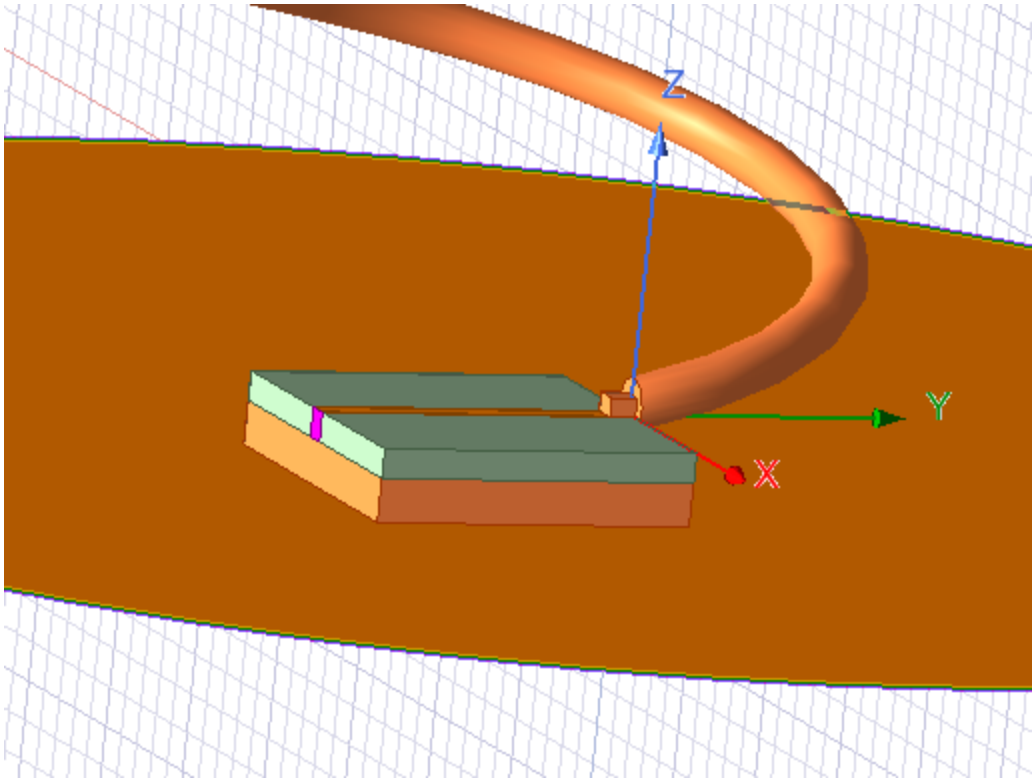


Gain is better

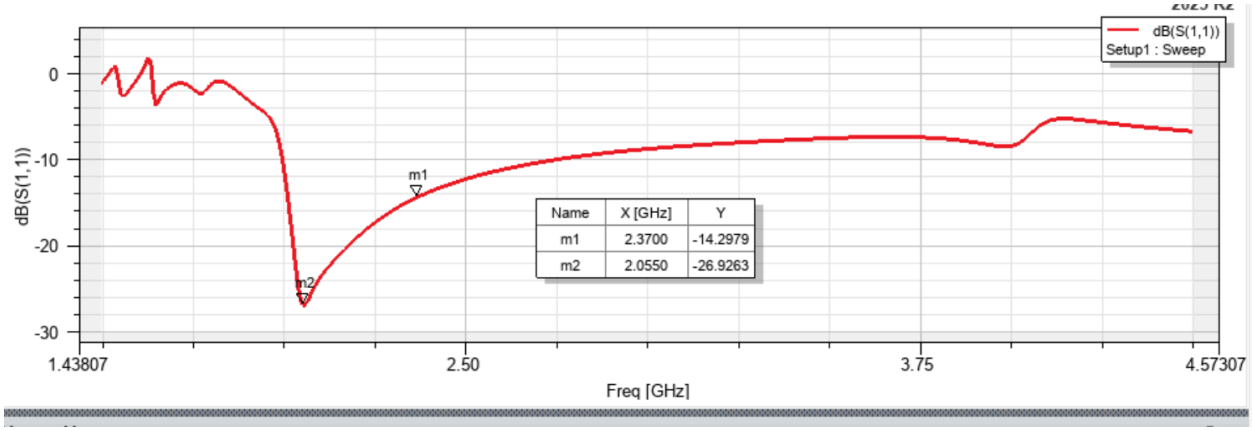
Impedance:


2025 R2
re(Z(1,1))
Setup1 : Sweep
Name
X [GHz]
Y
m1
2.4525
132.1302
re(Z(1,1))
2000
1000
0
m1
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Freq [GHz]


Z Parameter Plot2
ReDesign2
Ansys
2025 R2
im(Z(1,1))
Setup1 : Sweep
Name
X [GHz]
Y
m1
2.4600
-16.4783
im(Z(1,1))
500
0
-500
m1
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Freq [GHz]



Now testing with quarterwave matching track

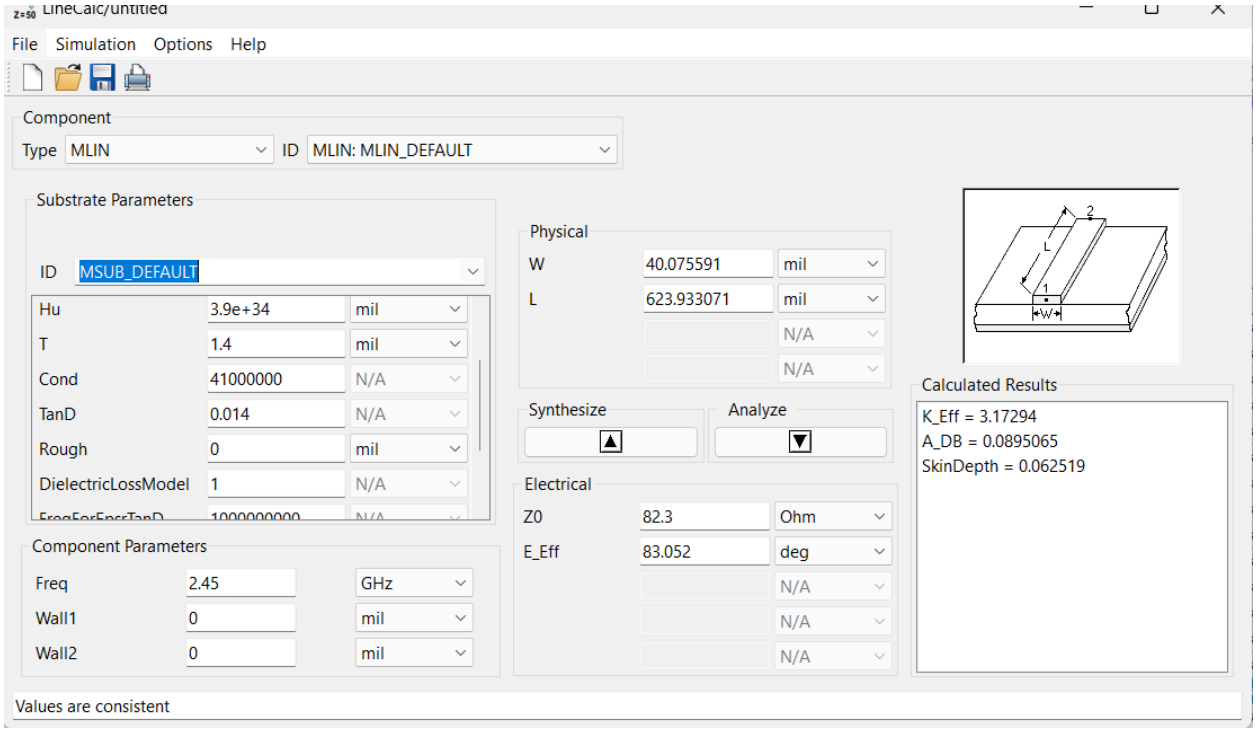


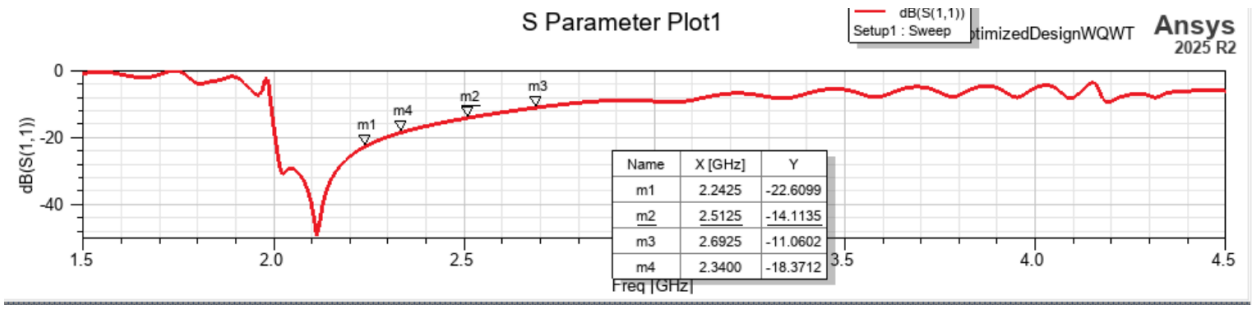
Does not work as expected, going to try matching track idea instead

$$= Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right).$$

Using the equation above and taking the really and imaginary load impedances plotted above.

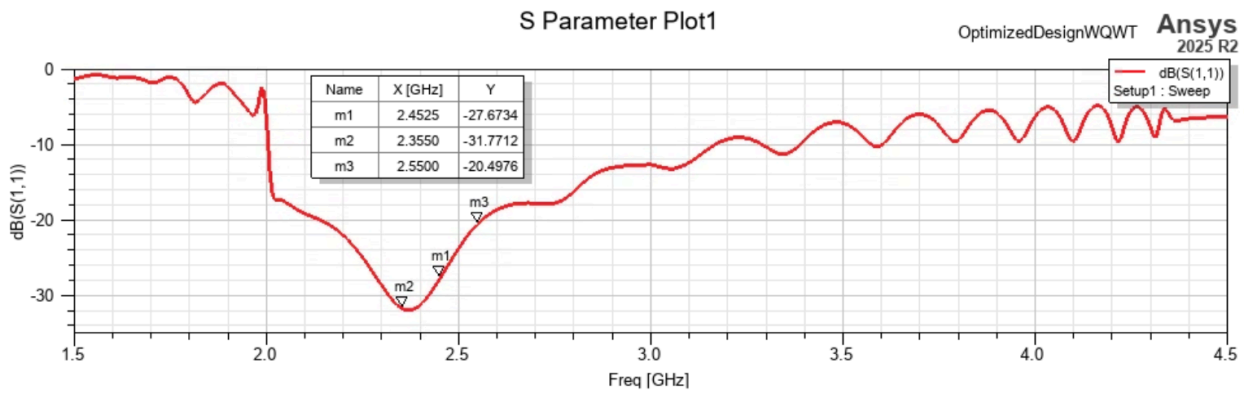
The target impedance is 82.3 ohms, and the length is .2307 lambda_0





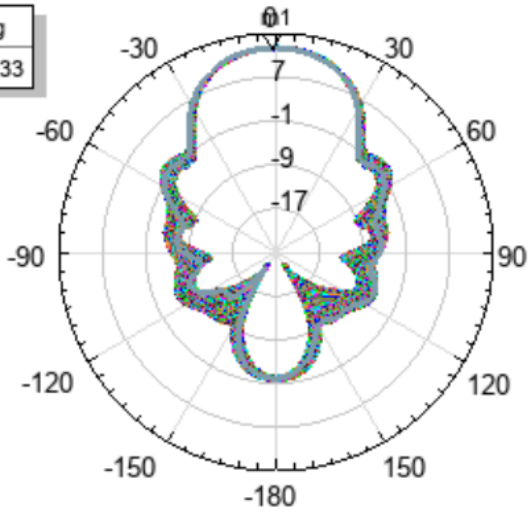
Almost really good have to change the match track a little bit,

Trying 70 degree phase shift, making line shorter.



LFGGGG

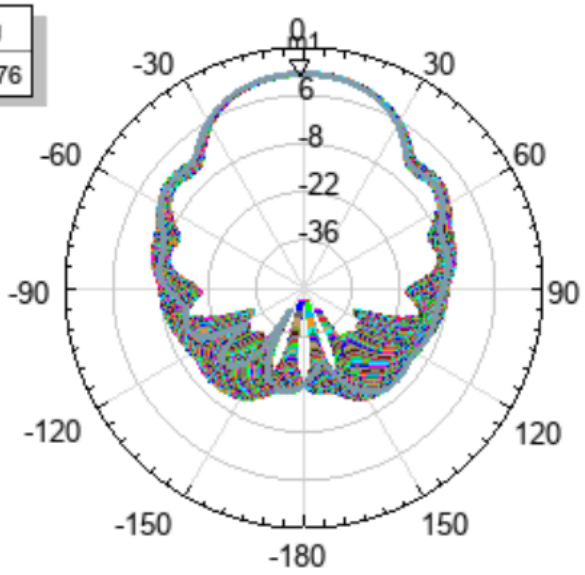
Name	Theta [deg]	Ang	Mag
m1	0E+00	4.9738E-13	12.3733



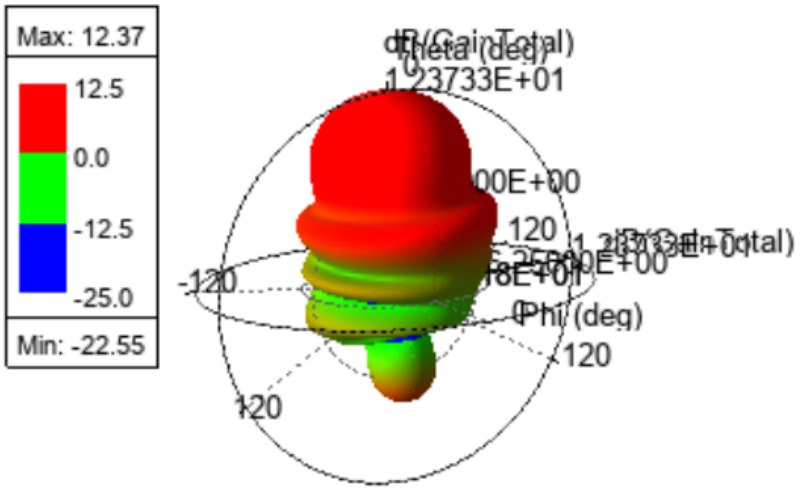
	Freq [GHz]	dB(PeakRealizedGain) Setup1 : LastAdaptive
1	2.400000	12.372331

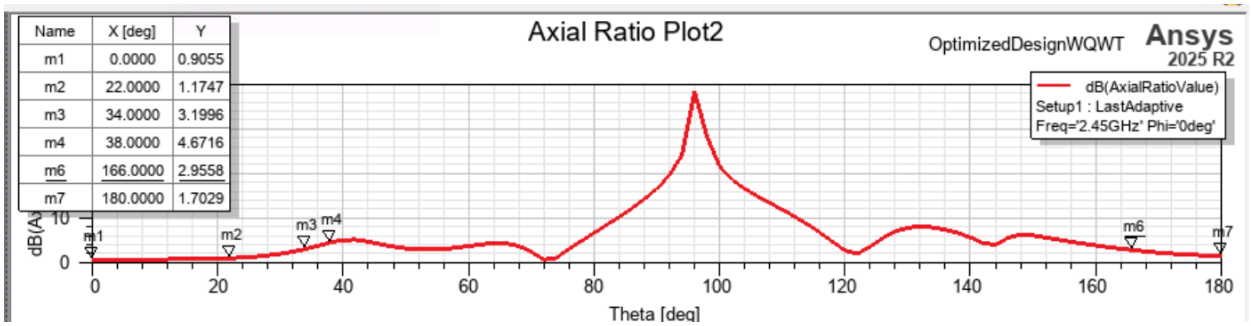
Gain Plot2

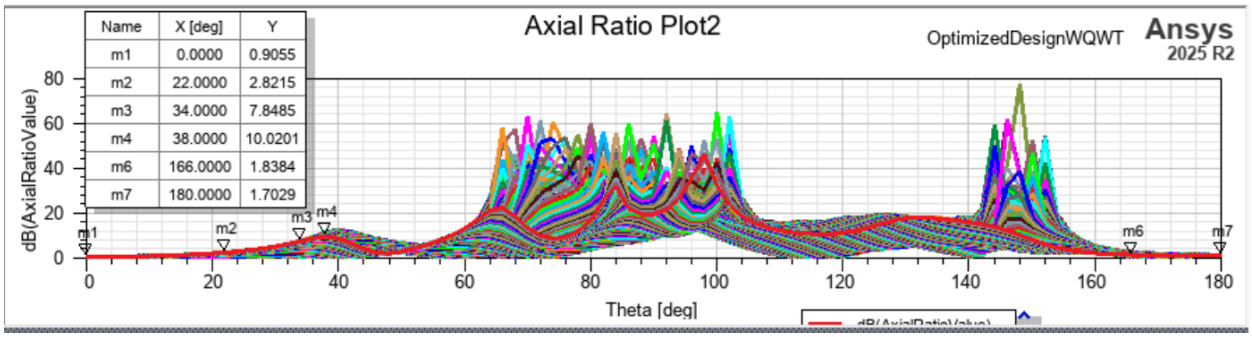
Name	Theta [deg]	Ang	Mag
m1	0E+00	4.9738E-13	12.3576



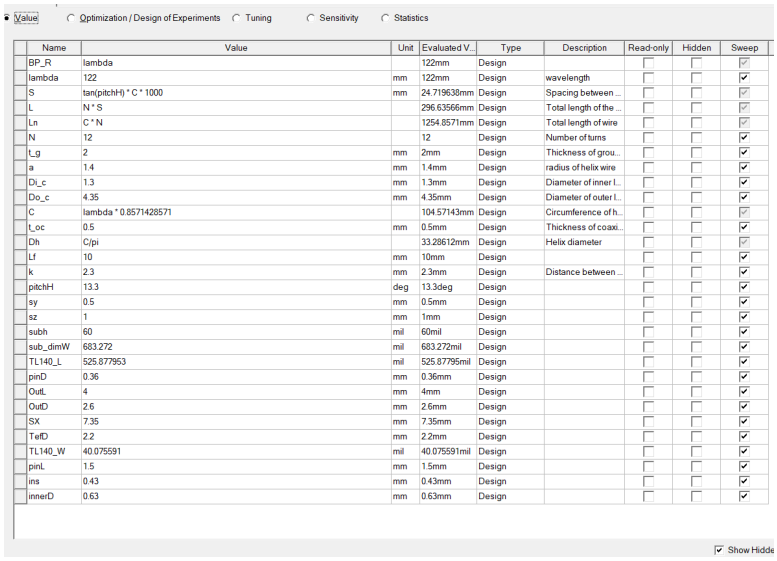
Gain Plot3




Axial Ratio Plot2
OptimizedDesignWQWT
Ansys
2025 R2
dB(AxialRatioValue)
Setup1 : LastAdaptive
Freq='2.45GHz' Phi='0deg'
Name | X [deg] | Y
m1 | 0.0000 | 0.9055
m2 | 22.0000 | 1.1747
m3 | 34.0000 | 3.1996
m4 | 38.0000 | 4.6716
m6 | 166.0000 | 2.9558
m7 | 180.0000 | 1.7029
dB(A
Theta [deg]


Axial Ratio Plot2
OptimizedDesignWQWT
Ansys
2025 R2
Name | X [deg] | Y
m1 | 0.0000 | 0.9055
m2 | 22.0000 | 2.8215
m3 | 34.0000 | 7.8485
m4 | 38.0000 | 10.0201
m6 | 166.0000 | 1.8384
m7 | 180.0000 | 1.7029
dB(AxialRatioValue)
Theta [deg]

Design Parameters:



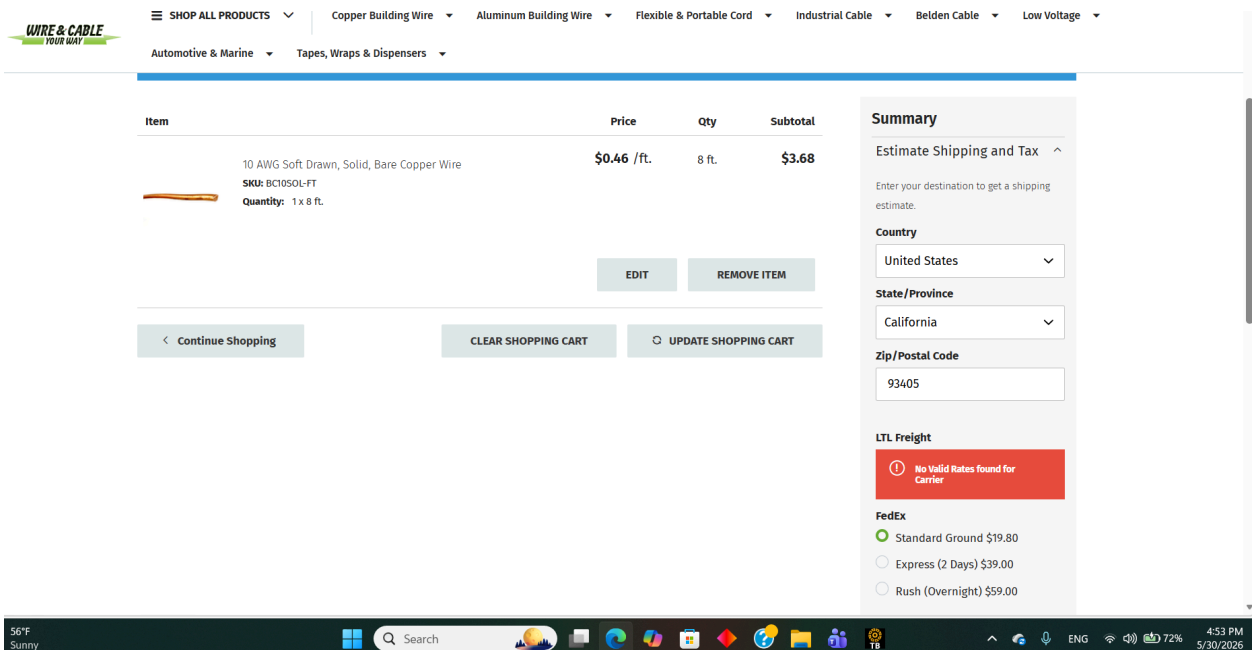
Value Optimization / Design of Experiments Tuning Sensitivity Statistics

Name	Value	Unit	Evaluated V...	Type	Description	Read-only	Hidden	Sweep
BP_R	lambda		122mm	Design		☐	☐	☑
lambda	122	mm	122mm	Design	wavelength	☐	☐	☑
S	tan(pitchH) * C * 1000	mm	24.719638mm	Design	Spacing between ...	☐	☐	☑
L	N * S		296.63566mm	Design	Total length of the ...	☐	☐	☑
Ln	C * N		1254.8571mm	Design	Total length of wire	☐	☐	☑
N	12		12	Design	Number of turns	☐	☐	☑
t_g	2	mm	2mm	Design	Thickness of grou...	☐	☐	☑
a	1.4	mm	1.4mm	Design	radius of helix wire	☐	☐	☑
Di_c	1.3	mm	1.3mm	Design	Diameter of inner l...	☐	☐	☑
Do_c	4.35	mm	4.35mm	Design	Diameter of outer l...	☐	☐	☑
C	lambda * 0.8571428571		104.57143mm	Design	Circumference of h...	☐	☐	☑
t_oc	0.5	mm	0.5mm	Design	Thickness of coaxi...	☐	☐	☑
Dh	C/pi		33.28612mm	Design	Helix diameter	☐	☐	☑
Lf	10	mm	10mm	Design		☐	☐	☑
k	2.3	mm	2.3mm	Design	Distance between ...	☐	☐	☑
pitchH	13.3	deg	13.3deg	Design		☐	☐	☑
sy	0.5	mm	0.5mm	Design		☐	☐	☑
sz	1	mm	1mm	Design		☐	☐	☑
subh	60	mil	60mil	Design		☐	☐	☑
sub_dimW	683.272	mil	683.272mil	Design		☐	☐	☑
TL140_L	525.877953	mil	525.87795mil	Design		☐	☐	☑
pinD	0.36	mm	0.36mm	Design		☐	☐	☑
OutL	4	mm	4mm	Design		☐	☐	☑
OutD	2.6	mm	2.6mm	Design		☐	☐	☑
SX	7.35	mm	7.35mm	Design		☐	☐	☑
TefD	2.2	mm	2.2mm	Design		☐	☐	☑
TL140_W	40.075591	mil	40.075591mil	Design		☐	☐	☑
pinL	1.5	mm	1.5mm	Design		☐	☐	☑
ins	0.43	mm	0.43mm	Design		☐	☐	☑
innerD	0.63	mm	0.63mm	Design		☐	☐	☑

Show Hidde

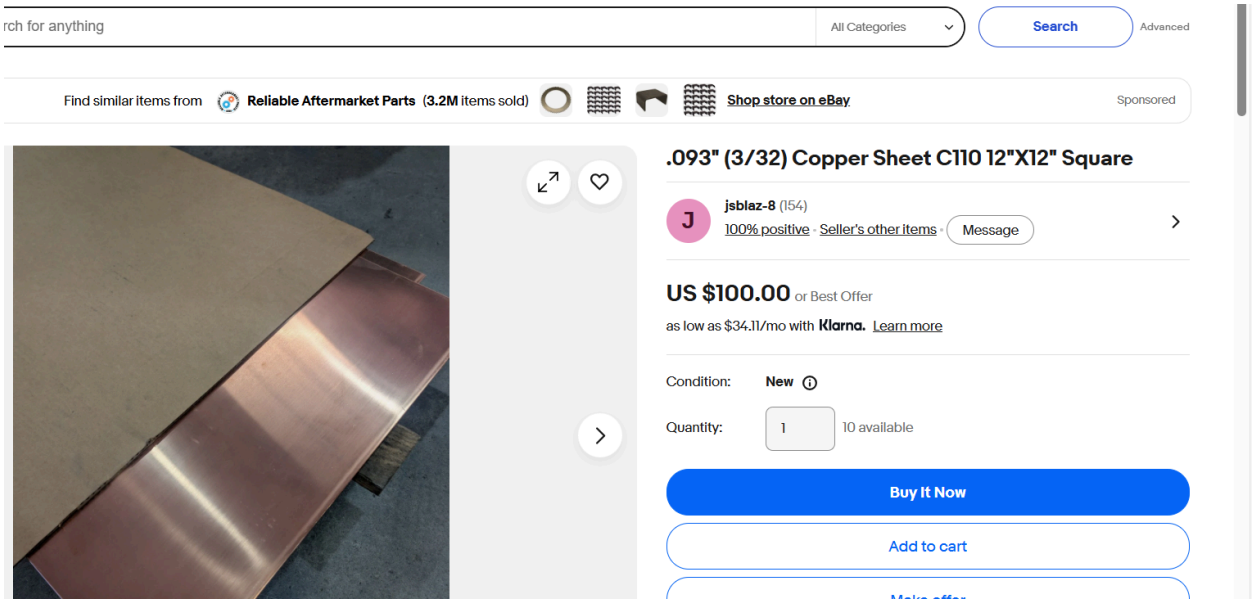
Bill of materials and where I got them from:

10 AWG Copper wire: [Shopping Cart]



Minimum $20 payment

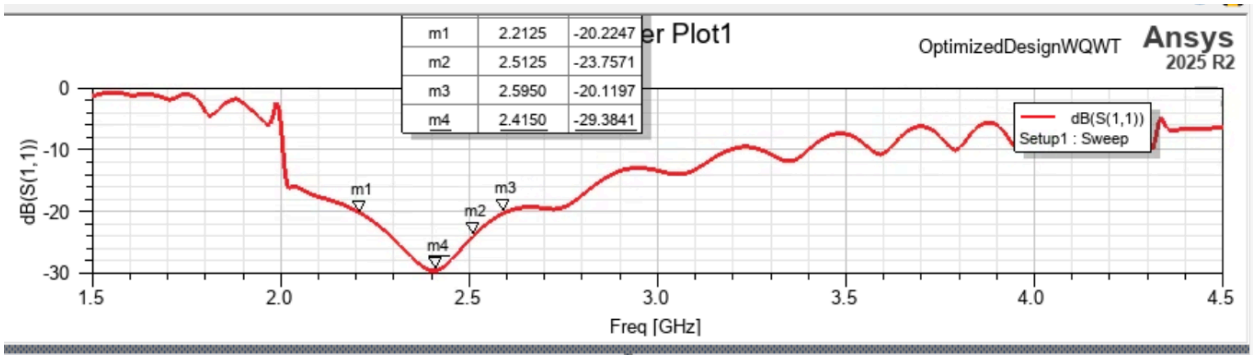
2.286 mm copper spacer:



bru this was basically the cheapest option

Going to go to home depot to buy the copper sheeting, going to use copper tape to bond ground plane of match track to back plane.

I



Still works with 10 AWG copper wire

Making the fixture compatible with the current design

With 12 awg wire

